Jaime Gómez

PhD Student in mathematics at EPFL
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Phone number: +41 78 243 38 67, **Citizenship:** Spanish



Education

EPFL — *Lausanne, Switzerland*
PhD studies in mathematics (dispersive PDEs and time-frequency analysis) — *2023–*

Supervisor: Prof. Joachim Krieger.
Research group: Partial Differential Equations.

ETH Zurich — *Zurich, Switzerland*
Master's degree in mathematics — *2021–2023*

Master's thesis supervisors: Prof. Alessio Figalli & Dr. João Pedro Gonçalves Ramos.
Thesis title: Stability of Faber-Krahn inequalities in time-frequency analysis.
Graduated with an ETH medal for an outstanding master thesis and a GPA of 5.84/6.

Universidad Autónoma de Madrid — *Madrid, Spain*
Bachelor's degree in mathematics with programming minor — *2017–2021*

Bachelor's thesis supervisor: Prof. Ana Vargas Rey.
Thesis title: The Fourier transform and Hausdorff measures. (ES)
Graduated with honors in my bachelor's thesis and a GPA of 8.77/10.

Languages and skills

Language skills

- **Spanish:** Native.
- **English:** Fluent.
- **Italian:** Intermediate.
- **French:** Basic.

Soft skills

- Analysis
- Problem solving
- Teamwork
- Organization

Musical and piano studies — *Salamanca, Spain*
Professional conservatory of Salamanca — *2009–2015*

Research

Stability of the Faber-Krahn inequality for the STFT — *Inventiones Mathematicae*
With André Guerra, João P.G. Ramos & Paolo Tilli. — *2024*

Uniform stability of concentration inequalities and applications — *Proc. London Math. Society*
With D. Kalaj, P. Melentijević & João P.G. Ramos. — *2025*

Multisoliton solutions and blow up for the L^2-critical Hartree equation — *ARMA*
With Tobias Schmid & Yutong Wu. — *2025*

Quantitative scattering estimate for the cubic Klein-Gordon equation — *In preparation*

With Joachim Krieger. *2026*

Sharp stability of the Faber-Krahn inequality for Bergman spaces and log-subharmonic functions *In preparation*
With D. Kalaj, P. Melentijević & João P.G. Ramos. *2026*

Refereed for: American Journal of Mathematics.

Awards and honors

2024 **Awarded** the ETH Medal for an outstanding master thesis.

2021 **Awarded** honors for my bachelor's thesis.

2017 **Awarded** the extraordinary high school award by the Castille and Leon autonomous region.

2017 **Awarded** for being among the best 100 grades in the selectivity exam in the university district of Salamanca.

2016 **5-th place** in the physics olympiad in Salamanca.

2016 **6-th place** in the mathematics olympiad in Salamanca.

Programming skills and projects

Programming skills

Programming minor (C, 1 yr), with mathematics-oriented courses (R, Matlab, 1/2 yr, SageMath, 1 yr).

- **Advanced:** C, Python, GDScript, LaTeX .
- **Basic:** C#, Swift, Matlab, HTML, CSS, JavaScript.

LaTeX2InD *Zurich, Switzerland*
Creator *2023*

Simulation-oriented *Madrid, Spain & Lausanne, Switzerland*
Projects in physics simulations and game development *2020-2025*

- **Godot:** Physics-based, interactive simulation of a moving sail boat, GDScript.
- **Unity**: Basic character controller, C#.

Interests

I enjoy learning languages (I am currently learning Italian) and I have a passion for traveling and photography. I thoroughly enjoy cooking, cinema, software design and aviation, and I also do various sports, mostly endurance-based, such as running and cycling (both road and gravel). I particularly enjoy the problem-solving aspects of climbing. In winter, skiing is my sport of choice. But most of all I love discovering new activities and places to explore while meeting people from all around the world.

Personal information

Citizenship: Spanish (European Union), with a **Swiss residence permit** (B).
Born in October 1999 in Salamanca, Spain.
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